

Deepak Mallampati

+1-330-554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |
Kent, OH (US-based, F-1 OPT, open to sponsorship)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) shipping production multimodal systems - four-provider AI vision integration in Manga Lens (live on Chrome Web Store), Stable Diffusion + ControlNet + LoRA + OpenPose conditioning in Pixel Character Synthesis Engine, YOLOv8 real-time detection with ~30% inference latency reduction, and transformer video summarization with 60-70% review-time savings. Production reliability via structured outputs, grounding, schema-validated multi-agent contracts, and FastAPI/Docker packaging with ~30% fewer post-deployment defects. Comfortable owning the full stack from inference plumbing through API surface.

SKILLS

Vision-Language Models • Multi-Provider VLM Integration • Structured Outputs & Grounding • Tool-Use Orchestration • Stable Diffusion / ControlNet / LoRA • FastAPI / Docker Inference • Agentic Workflows • Multimodal Production

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

LLM & GenAI: LLM application development, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search, responsible AI

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion, LoRA fine-tuning, video analytics, VLM provider integration

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, structured logging, observability

Domains: Multimodal Systems, Computer Vision, Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - USA (Healthcare Technology, AI Consulting)

- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by ~30%.
- Developed **agentic LLM workflows** with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~25%.
- Built **RAG systems** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings improved contextual retrieval precision ~35%, reduced irrelevant retrieval >30%, retrieval-grounded response alignment >90%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show, care engagement, support prioritization; recall +15-20% on high-risk categories at precision >90%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Hyderabad, India (Video Analytics, Applied NLP)

- Built a **transformer-based video summarization and highlight-generation system** across 5,000+ recorded sessions; manual review time reduced 60-70% (hours to <5 minutes per video); ~85% **alignment** with human-curated highlights.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~30% and raised contextual answer accuracy >85%.
- Deployed the stack as a **Flask-based API** with a lightweight web interface.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India

- Built **CI/CD pipelines with Jenkins**; deployment errors ~30%, release cycle time -35-40%.
- Designed performance dashboards using **SQL Server DMVs and Grafana**; cut incident recurrence ~25%.

PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation (shipped, Chrome Web Store)

Full-stack browser extension built independently - Manifest V3, content scripts, service workers, and `captureVisibleTab`. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local) with per-provider payload handling (WebP for cloud, JPEG for Ollama to avoid CUDA crash). Multi-section capture pipeline for tall webtoon panels via viewport-slice screenshots with coordinate remapping and dedup. Seven-day translation cache, per-domain selectors for 29 sites, privacy policy, narrowed host permissions. **Direct VLM-integration evidence.**

Agentic Pixel Character Synthesis & Animation Engine

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis with sprite-sheet export. **Stable Diffusion fine-tuning + LoRA** for identity consistency, **ControlNet + OpenPose/MediaPipe** for pose conditioning, latent-space consistency for animation smoothness, and an **LLM-based agent orchestrator** that decomposes high-level prompts into generation tasks. Backend: PyTorch, Hugging Face Diffusers, FastAPI, Docker.

Driver Drowsiness Detection System - Real-Time YOLOv8 Fatigue Monitoring

Replaced a two-stage CNN pipeline with a **unified YOLOv8** detection-and-classification model; reduced inference latency **~30%**. LabelImg annotations, augmentation for lighting/head-pose robustness, sliding-window confidence aggregation to suppress blink-driven false positives (**~25% reduction**), adaptive frame skipping, NMS tuning for stable real-time operation via OpenCV.

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation; approximate-nearest-neighbor similarity search for duplicate detection; explainable risk scoring with reasoning traces. Production-reliability pattern for tool-use orchestration.

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating multimodal APIs (Perplexity Sonar + Replicate image models). Introduced **intermediate structured prompt transformation layers** - improved contextual alignment **~30%** and first-pass image success rate **~25-30%**.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025